SYLLABUS

Data and the State: How Governments "See" People and Places

DATA 20195 / 30195
Moon Duchin, Spring 2026

1 Course Basics

Instructor & Lecture. This course is taught by Moon Duchin (mduchin@uchicago.edu), Professor of Data Science & Computer Science. Lectures are Monday/Wednesday 1:30-2:50. My main office is in DSI 308, but you can sometimes find me in the lab space for the Data and Democracy Research Initiative at JCL 375.

Teaching Assistants & Sections. Emma Ross (emross) and Michael Xiao (mx99) will run required discussion/lab sections on Fridays. One sections runs 1:30-2:50 and one runs 3-4:20.

Description. In order to regulate and govern, states must marshall data: the whos, whats, and wheres of a nation get rendered in records and statistics, at least approximately. One of the most fundamental instruments for the datafication of the population is a census – and the very first Article of the U.S. Constitution calls for one to be conducted every ten years. That same sentence is also the source of the notorious "Three-Fifths Compromise" establishing an enumeration rule where enslaved people count fractionally as much as free people: political arithmetic. In this course we will use Census data as a jumping-off point for an investigation of data practices of governance,looking at a host of frameworks and mechanisms to classify, measure, and locate. Students will learn to use a suite of modern packages for geospatial data science in Python, including Pandas, GeoPandas, Shapely, and Matplotlib, and readings will include texts from geography, anthropology, sociology, and science and technology studies. No prior knowledge assumed.

2 Assignments and Assessment

Each week we will be working through a Python notebook together (posted to the course GitHub) and you will be expected to do ≈ 30 pages of reading.

On a weekly basis, students will submit two items. First is a small data product (such as a plot or table) created during the Friday lab section and submitted at the end. This doubles as taking attendance for labs. Second is a weekly assignment in which students will be expected to briefly address the reading, and to creatively modify the code in the notebook, or write independent code, to produce a novel data product. Assignments are available Fridays, due the following Wednesday on gradescope, and there's the possibility of an automatic extension to Friday 1:30pm. You can use this option up to three times without needing to give any reason.

There are three short in-class assessments and one "micro-project." Projects will be short reports and visual data products written up solo but discussed with a small assigned group. Students can find a novel data source and curate it for use, or can use the datasets introduced in the class in a novel way. Projects must articulate an argument about the topic of choice, supported by data. Project topics should be approved by May 6; first draft due May 20; final submission date Weds May 27.

Academic integrity. Attribution statements are expected on homework assignments, documenting human or non-human assistance. We will be working collaboratively toward our AI policy over the course of the term. Assignments can be worked on in groups, but precise contents must be individual to the submitter.

Grading. For each prompt, work is typically graded on a four-point scale, where 3 out of 4 is good/solid work, and the score of 4/4 is reserved for exceptionally strong or creative work. At the end of the term, a score of 75% will earn an A-. Class participation is largely measured in lab sections, though participation in lecture is highly welcome. The grade breakdown is 10% participation, 30% weekly assignments, and 20% for each of three tests (undergrad) or two tests and a micro-project (grad).

Lab attendance is required. In the case of an absence, students are expected to fill out an Absence Form, typically by the following day, or as soon as possible in extraordinary circumstances.

3 Learning Outcomes

Goals of this course include the development of tools from data science and computing in order to tackle questions of social structure and culture, and to enable robust analysis and design of public policy. Skills to be developed include coding, statistical analysis, visualization/narrative-building, and critique. The core learning objectives are:

1. Write code to curate, clean, analyze, and visualize spatial data;
2. Gain an understanding of historical processes of classification and regulation;
3. Implement a descriptive or predictive analysis using appropriate data and statistical and/or computational methods;
4. Clearly communicate your process and results as a data narrative through visualizations, context, textual description, and oral presentation;
5. Identify limitations and potential flaws, and seek systematic improvements, in data, data-generating processes, and research methods.

4 Accommodations and Course Norms

I am committed to the accessibility of my courses for students from all backgrounds. For years I have worked to build courses in keeping with what is now called UDL, or *Universal Design for Learning.* This means that I will make the course meetings, assignments, and assessments as flexible and inclusive as possible. Anyone can access extra time on exams, and alternative formats can be proposed for assignments within the confines of administrability for a large course. Official accommodations can be registered through the office of Student Disability Services. Agreement about appropriate accommodations must be reached at least two weeks in advance of assignments or assessments for which an alternative structure is requested.

Laptop policy. If you want to use a laptop in class, I ask that you sit in the front row. For everyone else, please consider taking notes on paper! It is a different learning experience and you might like it. The reason for this policy is just to encourage more engagement and eye contact during class, though I understand if a laptop or other tech is best suited to your learning needs.

5 Topics and Schedule

Here is a plan for topic pacing, subject to change. The first-listed reading in each week is required, and the others are recommended.

Unit I: Introduction to Census and Geography

Week 1 Course Intro and Census I – why we count, who counts

Readings: James Scott, *Seeing Like a State*, Introduction

Melissa Nobles, *Shades of Citizenship*, Chapter 1: "Race, Censuses, and Citizenship"

Dataset: Decennial Census (PL94-171)

Week 2 Census and Geography – classifying people,

Readings: Ian Hacking, "Kinds of People: Moving Targets"

Ian Hacking, *The Taming of Chance*, Chapter 3, "Public Amateurs, Secret Bureaucrats"

B.R. Ambedkar, "From Millions to Fractions"

Dataset: American Community Survey (ACS)

Week 3 Geography II – units, hierarchies, contested boundaries

Readings: Mark Monmonier, *Drawing the Line*, Chapter 5, "Boundary Litigation and the Map as Evidence"

Dataset: Decennial Census and ACS, continued

—Assessment 1 (Weeks 1-3)—

Unit II: How Things Move

Week 4 Migration and relocation

Reading: Enrico Moretti, *The New Geography of Jobs*, Chapter 5, "The Inequality of Mobility and Cost of Living"

Dataset: ACS migration matrix and PUMS aggregated microdata

Week 5 Flows: Weapons, organs, votes

Readings: SIPRI background and position papers

Kieran Healy, *Last Best Gifts*, Chapter 1, "Exchange in Human Goods"

Jonathan Rodden, *The Long Shadow of the Industrial Revolution*, Intro–§1.2

Datasets: SIPRI database of international arms transfers

NY U.S. Senate election 2022; MA Presidential election 2012

STAR files with geodata supplemented by OPTN

Week 6 Cities and transit

Reading: Donald Shoup, *The High Cost of Free Parking*, Chapter 1, "The Twenty-first Century Parking Problem"

Dataset: NYC Open Data Portal and OSMnx for network structure

—Assessment 2 (Weeks 4-6)—

Unit III: Where We Live

Week 7 Policing and incarceration

Reading: Ruth Wilson Gilmore, *Golden Gulag*, Introduction, “Coercion”

Elizabeth Hinton, *From the War on Poverty to the War on Crime*, Chapter 8, “Crime Control as Urban Policy”

Dataset: arrest and incident data from Chicago and San Francisco, Vera Institute data on incarceration trends

Week 8 Housing and segregation

Readings: Richard Rothstein, *The Color of Law*, Chapter 1, “If San Francisco, Then Everywhere?”

Kenneth Jackson, *Crabgrass Frontier*, Intro/Chapter 1, or *Federal Subsidy and the American Dream* (article)

Datasets: University of Richmond redlining project

NYC Open Data housing data, Princeton Eviction Lab, Housing Data Coalition

Week 9 Neighborhoods, communities, districts

Readings: Jane Jacobs, *The Death and Life of Great American Cities*, Chapter 6, “The Uses of City Neighborhoods”

Garrett Nelson, *The Elusive Geography of Communities*

Datasets: NYT Upshot neighborhood project

Districtr community mapping portal

—Assessment 3 (Weeks 7-9)—